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Project: Rockbuster Stealth – Relational Database

Document Type: Data Dictionary (Final – Achievement 3)

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Description:

This document describes the structure of the Rockbuster Stealth relational database.

It lists all fact, dimension, and bridge tables used throughout Achievement 3,

including their columns, data types, and relationships (links to, links from, and unique keys).

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1. Overview of Schema

The Rockbuster database follows a **snowflake schema**:

- **Fact tables** store transactional or measurable data (payments, rentals, inventory).
- **Dimension tables** store descriptive attributes (customers, films, stores, locations, etc.).
- **Bridge tables** resolve many-to-many relationships (e.g. films and actors, films and categories).

Main relationships:

- Customers are linked to stores and addresses.
- Films are linked to languages and categories, and appear in inventory copies at stores.
- Rentals join customers, inventory items, and staff.
- Payments link back to rentals, customers, and staff.
- Locations are split across country → city → address.

2. Fact Tables

2.1 payment (Fact Table)

Stores individual payment transactions for rentals.

Columns:

Column	Data Type	Description
payment_id	integer	Primary key; unique ID for each payment
customer_id	smallint	FK → customer.customer_id
staff_id	smallint	FK → staff.staff_id
rental_id	integer	FK → rental.rental_id
amount	numeric(5,2)	Amount paid
payment_date	timestamp	Timestamp of payment

Links To:

- customer.customer_id
- staff.staff_id
- rental.rental_id

Links From:

- (No other tables link to payment; it is a terminal transaction table.)

Unique Keys:

- payment_id (Primary Key)

2.2 rental (Fact Table)

Represents each rental transaction of a physical film copy.

Columns:

Column	Data Type	Description
rental_id	integer	Primary key
rental_date	timestamp	Date/time rental began
inventory_id	integer	FK → inventory.inventory_id
customer_id	smallint	FK → customer.customer_id
return_date	timestamp	Date/time returned
staff_id	smallint	FK → staff.staff_id
last_update	timestamp	Timestamp of last update

Links To:

- inventory.inventory_id
- customer.customer_id
- staff.staff_id

Links From:

- payment.rental_id

Unique Keys:

- rental_id (Primary Key)

2.3 inventory (Fact Table)

Represents individual physical copies of films held at each store.

Columns:

Column	Data Type	Description
inventory_id	integer	Primary key
film_id	smallint	FK → film.film_id
store_id	smallint	FK → store.store_id
last_update	timestamp	Timestamp of last update

Links To:

- film.film_id
- store.store_id

Links From:

- rental.inventory_id

Unique Keys:

- inventory_id (Primary Key)

3. Bridge Tables

3.1 film_actor (Bridge Table)

Resolves the many-to-many relationship between films and actors.

Columns:

Column	Data Type	Description
actor_id	smallint	FK → actor.actor_id
film_id	smallint	FK → film.film_id
last_update	timestamp	Timestamp of last update

Links To:

- actor.actor_id
- film.film_id

Links From:

- (No further tables link to this; it is used in queries to join films and actors.)

Unique Keys:

- Composite key: (actor_id, film_id)

3.2 film_category (Bridge Table)

Resolves the many-to-many relationship between films and categories (genres).

Columns:

Column	Data Type	Description
film_id	smallint	FK → film.film_id
category_id	smallint	FK → category.category_id
last_update	timestamp	Timestamp of last update

Links To:

- film.film_id
- category.category_id

Links From:

- (No further tables link to this; it is used in queries to join films and categories.)

Unique Keys:

- Composite key: (film_id, category_id)

4. Dimension Tables

4.1 actor (Dimension Table)

Stores information about actors who appear in Rockbuster films.

Columns:

Column	Data Type	Description
actor_id	serial	Primary key
first_name	character varying(45)	Actor first name
last_name	character varying(45)	Actor last name
last_update	timestamp	Timestamp of last update

Links To:

- (None; actor does not reference other tables.)

Links From:

- film_actor.actor_id

Unique Keys:

- actor_id (Primary Key)

4.2 address (Dimension Table)

Stores physical addresses for customers, staff, and stores.

Columns:

Column	Data Type	Description
address_id	serial	Primary key
address	character varying(50)	Street address
address2	character varying(50)	Additional address line (optional)
district	character varying(20)	District / region
city_id	smallint	FK → city.city_id
postal_code	character varying(10)	Postal / ZIP code
phone	character varying(20)	Phone number
last_update	timestamp	Timestamp of last update

- postal_code – character varying(10) – Postal or ZIP code.
- phone – character varying(20) – Phone number associated with the address.
- last_update – timestamp – Timestamp of last update.

Links To:

- city.city_id

Links From:

- customer.address_id
- staff.address_id
- store.address_id

Unique Keys:

- address_id (Primary Key)

4.3 category (Dimension Table)

Defines film categories (genres).

Columns:

Column	Data Type	Description
category_id	serial	Primary key
name	text	Category name (genre)
last_update	timestamp	Timestamp of last update

Links To:

- (None; category does not reference other tables.)

Links From:

- film_category.category_id

Unique Keys:

- category_id (Primary Key)

4.4 city (Dimension Table)

Stores city information.

Columns:

Column	Data Type	Description
city_id	serial	Primary key
city	character varying(50)	City name
country_id	smallint	FK → country.country_id
last_update	timestamp	Timestamp of last update

Links To:

- country.country_id

Links From:

- address.city_id

Unique Keys:

- city_id (Primary Key)

4.5 country (Dimension Table)

Stores country information.

Columns:

Column	Data Type	Description
country_id	serial	Primary key
country	character varying(50)	Country name
last_update	timestamp	Timestamp of last update

Links To:

- (None; country does not reference other tables.)

Links From:

- city.country_id

Unique Keys:

- country_id (Primary Key)

4.6 customer (Dimension Table)

Stores information about Rockbuster customers.

Columns:

Column	Data Type	Description
customer_id	serial	Primary key
store_id	smallint	FK → store.store_id
first_name	character varying(45)	Customer first name
last_name	character varying(45)	Customer last name
email	character varying(50)	Customer email
address_id	smallint	FK → address.address_id
activebool	boolean	Whether account is active (boolean flag)
create_date	date	Account creation date
last_update	timestamp	Timestamp of last update
active	integer	Active status flag (numeric)

Links To:

- store.store_id
- address.address_id

Links From:

- rental.customer_id
- payment.customer_id

Unique Keys:

- customer_id (Primary Key)

4.7 film (Dimension Table)

Contains information about films in the Rockbuster catalog.

Columns:

Column	Data Type	Description
film_id	serial	Primary key
title	character varying(255)	Film title
description	text	Film summary / description
release_year	year	Year of release
language_id	smallint	FK → language.language_id
rental_duration	smallint	Rental duration in days
rental_rate	numeric(4,2)	Rental price
length	smallint	Film length in minutes
replacement_cost	numeric(5,2)	Replacement cost
rating	mpaa_rating	MPAA rating (G, PG, PG-13, etc.)
last_update	timestamp	Timestamp of last update
special_features	text[]	List of special features
fulltext	tsvector	Full-text search vector

Links To:

- language.language_id

Links From:

- inventory.film_id
- film_actor.film_id
- film_category.film_id

Unique Keys:

- film_id (Primary Key)

4.8 language (Dimension Table)

Defines languages used in the film catalog.

Columns:

Column	Data Type	Description
language_id	serial	Primary key
name	character varying(20)	Language name
last_update	timestamp	Timestamp of last update

Links To:

- (None; language does not reference other tables.)

Links From:

- film.language_id

Unique Keys:

- language_id (Primary Key)

4.9 staff (Dimension Table)

Stores information about Rockbuster staff members.

Columns:

Column	Data Type	Description
staff_id	serial	Primary key
first_name	character varying(45)	Staff first name
last_name	character varying(45)	Staff last name
address_id	smallint	FK → address.address_id
picture	bytea	Staff picture
email	character varying(50)	Staff email
store_id	smallint	FK → store.store_id
active	boolean	Active status
username	character varying(16)	Staff username
password	character varying(40)	Staff password
last_update	timestamp	Timestamp of last update

Links To:

- address.address_id
- store.store_id

Links From:

- store.manager_staff_id
- rental.staff_id
- payment.staff_id

Unique Keys:

- staff_id (Primary Key)

4.10 store (Dimension Table)

Stores information about Rockbuster stores.

Columns:

Column	Data Type	Description
store_id	serial	Primary key
manager_staff_id	smallint	FK → staff.staff_id (store manager)
address_id	smallint	FK → address.address_id
last_update	timestamp	Timestamp of last update

Links To:

- staff.staff_id (manager_staff_id)
- address.address_id

Links From:

- customer.store_id
- inventory.store_id
- staff.store_id

Unique Keys:

- store_id (Primary Key)